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RESEARCH ARTICLE

Long Trip Charging Planning of Battery Electric Vehicle Considering Vehicle Waiting Time Forecast at Fast Charging Stations: A Mixed-Integer Dynamic Programming Approach

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ABSTRACT Battery electric vehicles have shared concerns such as range anxiety and long charging times compared to conventional vehicles. To consider an electric vehicle as a replacement for a conventional vehicle, it is essential to reduce the cost of ownership and encourage widespread adoption without range concerns. It is appropriate that charging planning should be utilized to tackle such problems. It can also be helpful to have information from charging planning services to apprise incoming vehicles for charging to prepare their trip itinerary. For long-distance road trips, choosing appropriate charging stations can be instrumental in achieving the optimization goal. Selecting the right charging station can affect the total trip time. To optimize charging planning, we utilize mixed-integer dynamic programming to achieve the global optimal solution. The information on vehicle waiting time is used and parameterized from open-source US household travel survey information. Probability distributions of daily trips, daily mileage, and probability distribution of arrival vehicle SOC can be derived. Monte Carlo simulation generates the waiting time at the charging station based on the probability distributions and station configuration. The presented optimal charging plan method is validated by multiple test cases. The test cases are designed based on US Alternate Fuel Corridors utilizing interstate highways. For the selected test scenarios up to 9.6% of time saving is observed when there is no waiting time at the charging stations. The validation tests with waiting time encountered at the charging station, time saving of up to 15.1% is achieved with energy level closer to the minimum limit at the end of the trip. This charging planning method significantly enhances the viability of EVs for long-distance travel by addressing core concerns like range anxiety and lengthy trip time.

INDEX TERMS Long trip charging planning, mixed integer optimization, battery electric vehicles, connected vehicles.

I. INTRODUCTION

Battery electric vehicles (BEVs) are the widely accepted solution for stringent emissions norms and mitigating drawbacks of conventional vehicles, such as tailpipe emissions [1]. The annual electric vehicle sales are estimated to touch 22 million by 2025 [2]. By 2035, it is indicated that 100 million BEVs will be on the road worldwide [2]. Technological developments in battery cells and battery packs have bolstered the large-scale production of BEVs possible [3], [4]. Among all the different chemistry of battery cells, lithium-ion batteries have been widely adopted [5].

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However, BEVs have shared concerns such as range anxiety and long charging times compared to conventional vehicles. It is considered that charging planning can be utilized to tackle such problems. Charging during the trip may be necessary for long-distance commuting and road trips. In such cases, choosing an appropriate charging station can be instrumental in achieving the optimization goal of minimum trip time and residual energy at the destination. Selecting the right charging station based on its location and charging rate can affect their charging time and the total trip time. The decision points on a trip can thus be the change in charging station location and the extent of charging to be done at the chosen charging location or locations.

There are different levels of chargers. AC Level 1, AC Level 2, DC Fast Charging (DCFC), and Extreme Fast Charging (XFC) [6]. AC Level 1 & 2 are predominantly the home chargers for overnight charging. The charging power of level 1 is between 1.4 to 1.9 kW with a charging voltage of 110/120 V AC. The amount of throughput is the lowest among the chargers. The charging time can be up to 36 hours, considering 200 miles of charge. Level 2 has a moderate power throughput of 3.1 to 19.2 kW with a 208/240 V AC electric circuit. The charging time can range from 7 to 8 hours for 200 miles. DCFCs are available in public or commercial charging stations, which have 440/480 V AC circuits converted to DC for EVs. Power levels can range from 50 kW to 350 kW [6]. DCFC can charge a vehicle with 200 miles range in an hour. XFCs are currently sparsely available in public or private charging stations but are prominent in the future of infrastructure development plans for large-scale EV adoption by the public. It would be possible to charge for 200 miles in 7.5 minutes [6], [7]. In [8] and [9] the authors share the study of EV fast charging on the battery degradation. They conclude, based on the study of 12,500 Tesla vehicles, that the manufacturers have made sufficient engineering solutions to consider frequent use of DC fast charging and its effects. Thus restricting the damage to a similar level of regular AC level charging. The authors in the previous article [6] also propose an optimal strategy for reducing cost and meeting power demand without straining a distribution network for XFC stations integrated with energy storage systems and PV solar systems.

Alternate Fuels Data Center [10] has information regarding charging station locations in the US and Canada. Current and upcoming infrastructure development bolsters the charging stations' availability on major routes. For example, on interstate freeways, DC fast charging stations are planned every 50 miles and within a 1-mile radius detour away from the route. The charging stations at different locations will serve the local population and long-travelling visitors. This adds to the complexity of waiting time due to arrival rate changes at different times of the day. It is essential to consider that the number of chargers in a station is limited. Thus, choosing a charging station during the trip is critical considering some delays due to rush hour congestion.

Range anxiety is a common concern among early EV adopters as expressed by Ford CEO [11], [12]. This is especially true for long-distance travellers with lower-cost BEVs that have smaller EPA estimated driving ranges. Range anxiety of BEV is handled by Thorgeirsson [13] through the evaluation of a different system architecture for the fastest route possible. Authors argue that the estimation based on Machine Learning with suggested system architecture can help reduce anxiety and help charging planning. Authors in [14] argue that the proposed methodology can increase revenue, handle power burden better, and queue concerns at the stations. Data-driven demand-based energy planning for electric vehicles with access available to Fast Charging Stations, AC level charging, and battery swapping service. Such a solution is recommended by authors for energy planning of electric vehicles; however, United States alternate fuel data centers do not share the information of existing battery swapping centers. This prohibits the implementation of the recommended solution in the United States. Forger et al. consider a capacitated charging station's availability for an electric vehicle energy planning problem [15]. A mixed integer linear programming is used to generate the initial policy which will be assembled with more complex conditions to develop a more detailed and comprehensive solution.

Energy planning based on route selection has been widespread. In [16], the authors discussed long-term energy planning for electric vehicles designated to complete assigned tasks. Researchers have incorporated an energy consumption estimation model in the proposed 2-stage strategy for energy consumption and routing problems. Another variation of task planning [17] uses graph theory and mixed integer programming to optimize an EV for daily activities. Efficient route planning based on topography and battery lifecycle is considered by Perger and Auer [18]. A multi-objective optimization with different weights is explored to find results for different preferences. A fleet energy planning system [19] is developed involving waiting time at charging stations to complete tasks. Multiple adapting routing and recharging for electric vehicles using heuristic methods are developed by Sweda et al. [20]. The authors assume a rectangular grid kind of network of charging stations which can be used to plan routing and charging policies. However, considering charging stations as grid handicaps real-world applications where the location of charging stations is not necessarily planned like a grid.

There is another approach to saving energy in the transportation sector using connected and automated vehicle (CAV) environment. In CAV environment, recent infrastructural developments in civil transportation technologies, such as automation of traffic lights and communication technologies, are available for the vehicles to benefit from. CAV benefits can be employed for reducing energy consumption in conventional, HEV/BEV and Fuel Cell vehicles. There can be long-term suitable information and short-term suitable information availability by deploying CAV technologies.

In the article [21], authors have shared strategies to utilize topographical information for energy saving through eco-driving. Short-term velocity profile generation is shared for plug-in hybrid electric vehicle (PHEV), 2nd Generation GM Volt, using quadratic programming [22]. Real-time suitable velocity prediction for MPC-based energy management of PHEV using the ACADO tool, which minimizes energy consumption, is discussed in [23] and [24]. A multi-stage optimization framework using short-term information for PHEV energy management is developed in [25], [26]. Authors utilize MPC for optimizing power-split using velocity profile for 2nd Gen. GM Volt PHEV. In the case of actuated and semi-actuated traffic intersections, the complexity increases, and a solution is proposed in [27]. Intelligent parking lot management to optimize revenue generation and vehicle charging needs during parking has been explored in [28]. The authors have utilized linear programming which has the potential to achieve faster convergence and maximize the profit for the parking lot.

All the existing literature treats the BEV energy and charging planning problem from the perspective of the routing problem. Many articles do not consider vehicle dynamics and powertrain constraints. In some papers, real world non-linear battery charging power characteristics are not considered. Our contributions through this research are: *i*) the proposed approach considers charging stations waiting time based on open-source information such as the National Household Travel Survey, daily mileage for surveyed population, and battery characteristics of the vehicle which affect the charging time, *ii*) the presented method is a global optimal solution.

The rest of the chapter is organized as follows: section II introduces monte-carlo (MC) simulation to estimate the probability distribution of arrival time and waiting time of BEV vehicles at charging stations and state of charge (SOC). Section III describes optimization formulation and mixed integer dynamic programming approach to select charging station/s and decide the limit of charging done at the charging stations. Section IV presents the validation of the optimization algorithm and discusses the results. Section V summarizes the presented work and recommends future work.

II. MONTE-CARLO SIMULATION FOR THE ESTIMATION OF VEHICLE ARRIVAL TIME AND WAITING TIME AT CHARGING STATIONS

In this section, we discuss the formulation of monte-carlo simulation to find the vehicle arrival time for charging at the charging station. In subsection A, vehicle arrival time is computed using various input parameters such as vehicle daily mileage, home departure SOC of vehicles, hourly daily trip distribution and arrival SOC threshold of vehicles at the charging station.

In subsection B, the monte-carlo simulation to compute the vehicle waiting time distribution is formulated utilizing the arrival time of vehicles at the charging station, along with other inputs such as charging port availability, arrival SOC of vehicles, individual vehicle models and their unique battery charging characteristics.

A. MONTE-CARLO SIMULATION FOR VEHICLE ARRIVAL TIME AT CHARGING STATION

In the consumer behavior report [29], the study shares that approximate 50% of BEVs are charged at home, and the remaining vehicles are charged somewhere else. On this basis, we assume that the mean of the SOC normal distribution for half of the vehicles leaving charged from home is 80%, with a standard deviation of 3.2%. The remaining vehicles that may be charged somewhere else will have 30% as the mean and a standard deviation of 3.2%. The standard deviation of 3.2 provide a reasonable spread, as shown in Figure 1. Home departure SOC is denoted as $\widehat{SOC}_{initial}$.

We assume that the arrival SOC threshold limit distribution at the fast-charging station has a mean of 25%, which is denoted as S_{SOC}^{BEV}. The standard deviation of arrival SOC threshold is selected to concentrate the population more on the mean value of 25% as the owners will not allow discharge to go towards dangerous levels when deciding to look for fast chargers. Charging vehicles when SOC is closer to 50% is considered contradictory to common knowledge. Home departure time has been studied and suggested for load profile modeling of electric vehicles in [30]. The information shared regarding probability density function for weekday home departure time is shown in [30] table 6, which is used to generate home departure time samples for 10000 vehicles in this study.

The probability distribution of the total trip distance travelled per day is obtained from the National Household Travel Survey (NHTS). The NHTS is an open database developed by the Oak-Ridge National Laboratory (ORNL) and funded by the Federal Highway Administration (FHWA). The survey provides data regarding the driving habits of survey participants. It is important to be aware that these survey data may be biased based on the population which has participated in the survey. In addition, the driving habits can change over time, which may change the survey results. Also, BEV early adopters' driving habits can be different compared to cost sensitive car buyers and driving habits. Readers who are interested in learning the limitations and bias embedded in NHTS survey data can refer to [31], [32], and [33].

The authors in [31] utilize the NHTS data and process it by removing duplicate, outlier, and incomplete data points. Figure 3 in the article [31] shows the probability distribution of the total trip distance travelled per day respectively. In this study, Figure 3 from the article [31] is used to generate 10000 R_n random samples, where R_n is the randomly generated cumulative daily trip mileage of the nth vehicle.

The daily driving miles of the vehicle majorly range from 5 to 40 miles a day [31]. Figure 3 in the article [32] and Table 2 in the article [33] show the probability distribution of

daily trip in terms of the time of the day, represented as λ. To find the arrival time at the charging station, we consider that the vehicles will go for charging as soon as the vehicle SOC reaches the arrival SOC threshold limit. Equation (1) defines the SOC declining based on λ, R_n and R_{max}. The hourly probability of the daily trip is referred to as λ. We assume λ to be constant for the hour. To convert λ into minute resolution the hourly value is divided by 60. So, at time t minutes, the probability is denoted as $\lambda(t)$. R_{max} is the maximum range of each BEV of the randomly generated 10000 samples of vehicles. R_n is the randomly generated cumulative daily trip mileage of the nth vehicle. R_{max} samples are based on the percentage distribution of samples shared in Table 1. Table 2 lists the parameters used for monte-carlo simulation. In equation (1), Δt is 1 minute and $\widehat{SOC}_{initial}$ is used as initial value for $\widehat{SOC}_n$ which updated at every time instance.

$$\widehat{SOC_n}(t+\Delta t) = \widehat{SOC_n}(t) - \left(\lambda(t) * \frac{R_n}{R_{max,n}} * 100\right) \quad (1)$$

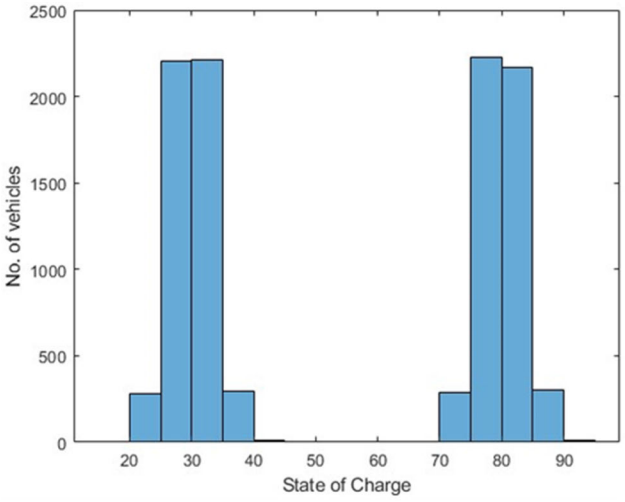


FIGURE 1. State of charge distribution for departing vehicles from home.

A monte-carlo simulation for 10000 vehicles is designed and executed to find the probability of vehicle arrival time at the fast-charging station. For the vehicles on the road, the state of charge is updated using the equation (1) at every time instance. Figure 2 shares the algorithm for generating the probability of vehicle arrival time at the charging station; henceforth, we identify the algorithm as AVAT in short form.

B. MONTE-CARLO SIMULATION FOR VEHICLE WAITING TIME AT CHARGING STATION

The BEV population consists of products from different manufacturers. Different manufacturers have built vehicles with different battery capacities. Based on the details published in [34] and consideration that 50% of vehicle owners charge at home and elsewhere. It is considered one FCS charging port for 15 vehicles. Table 1 shows the BEV models considered in monte-carlo simulation, their battery capacity, and the percentage share in samples. The battery charging power characteristics for each vehicle type are shown in Figure 3. The non-linear battery charging power characteristics make it difficult to estimate the charging time.

TABLE 1. Vehicle models and their battery capacities.

BEV Model	Battery Size (kWh)	Maximum Range, R_{max}(miles)	% Share in samples
Porsche Taycan	79.2 kWh	200	5
Tesla Model 3LR	82 kWh	353	30
Audi E-Tron	95 kWh	222	25
VW ID.4	82 kWh	251	20
Hyundai Kona	67.5 kWh	258	10
GM Bolt	60 kWh	238	10

TABLE 2. Parameters for monte-carlo simulation of arrival time probability.

Parameters	Description
R_n	Daily mileage sample
S_{SOC}^{BEV}	Arrival SOC threshold limit sample at charging station
R_{max}	Maximum vehicle range
$\widehat{SOC}_{initial}$	Home departure initial SOC sample
λ	Hourly probability of the vehicle daily trip

Figure 3 shows the charging curves of various types of BEVs [6], [35]. The battery characteristics information used in this work is the data recorded by Fastned, which operates charging stations. When vehicles stop for charging at their charging stations, the respective vehicle battery characteristics are recorded. Their policies related to data collection and information sharing are publicly available on the website [35]. We use monotone piecewise bicubic interpolation [36], implemented by 'pchip' function in Matlab to interpolate battery characteristics data from [35]. The BEV charging power depends on the power rate of the charging port and battery pack charging power characteristics. To explain with an example, consider a BEV with a maximum charge rate of 50 kW, which can be connected to a 300 kW charger. However, the vehicle will be charged at its own defined rate of 50 kW. The extreme fast charging capabilities may not be fully exploited due to vehicle battery characteristics limitations.

For computing the waiting time, we consider the probability distribution of vehicle arrival time at the charging station discussed in section II-A. We assumed that there would be 8 connecting ports at a charging station. It is considered that the maximum waiting time for a vehicle

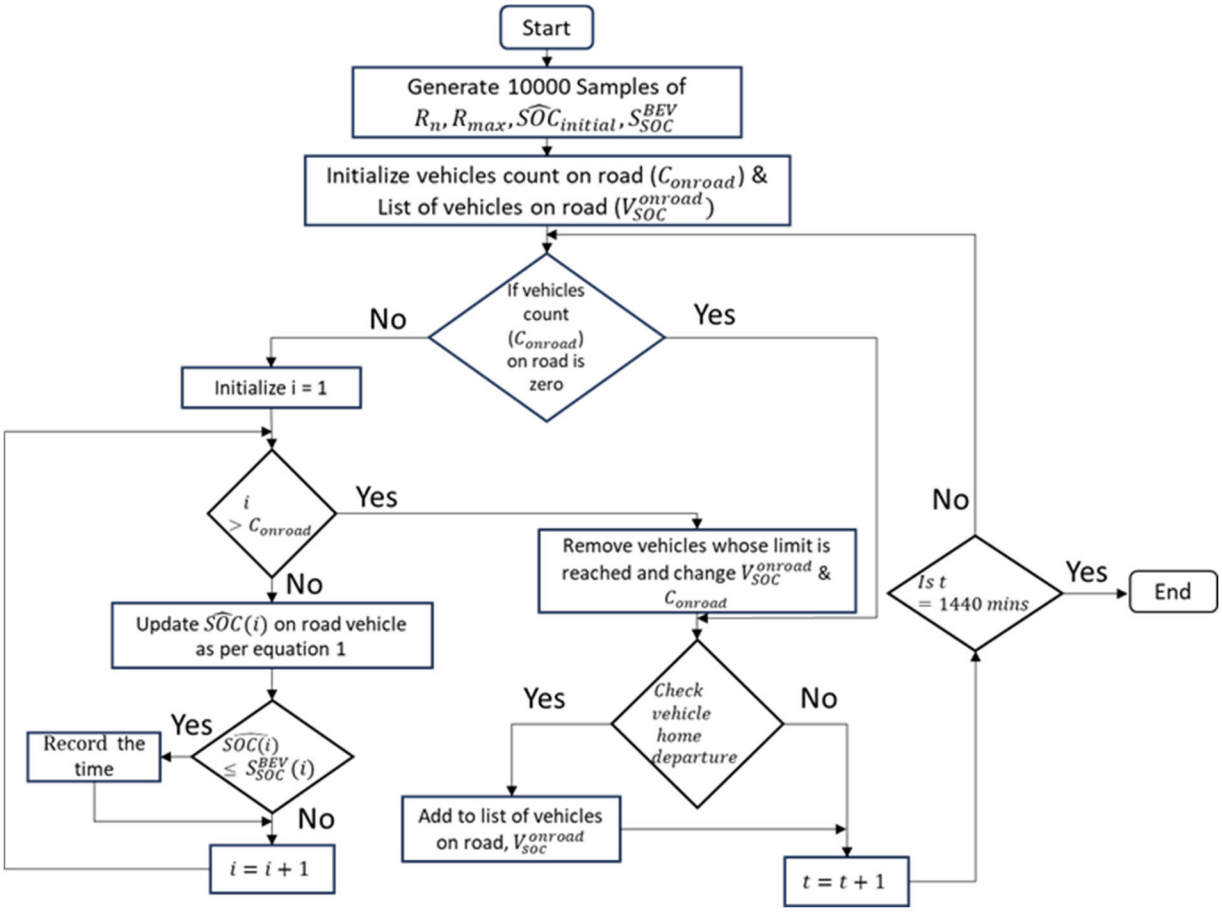


FIGURE 2. Flowchart of algorithm for vehicle arrival time.

is 15 minutes [37]. The vehicle will leave to find another charging station after waiting for 15 minutes. The simulation is based on the principle that the vehicle connected for charging will be charged to 90% of the battery capacity of the vehicle. The charging power during the charging follows the power charging curve for a respective battery pack. The update of SOC for every instance is based on equation (2).

$$SOC_{c,n}^{BEV}(t + \Delta t) = SOC_{c,n}^{BEV}(t) + \frac{(P_n^{BEV} * \eta * \Delta t)}{(C_n^{BEV})} * 100 \tag{2}$$

where $SOC_{c,n}^{BEV}$ represents the SOC of nth BEV connected for charging, P_n^{BEV} is the power charging rate for the nth vehicle, η is the efficiency of charging considered as 0.92, Δt is the incremental time instance which is 1 min in this case, and C_n^{BEV} is the battery capacity of the nth vehicle. Figure 4 provides a flowchart for the simulation algorithm to forecast the waiting time at each instance for a charging station. The process includes finding the list of vehicles that arrived at the station for each instance of time. If available ports for charging are less compared to the arrivals, then these vehicles are added to a waiting list. The individual waiting times of vehicles on the waiting list are monitored. When the

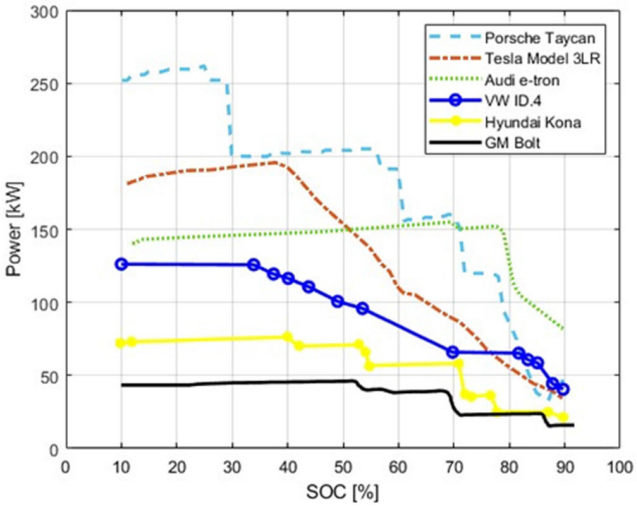


FIGURE 3. Different BEV's charging curves.

waiting time of a vehicle is greater than the maximum waiting time, then the vehicle is removed from the waiting queue assuming that this vehicle will leave to find another charging station. There is a separate charging list for the vehicles being

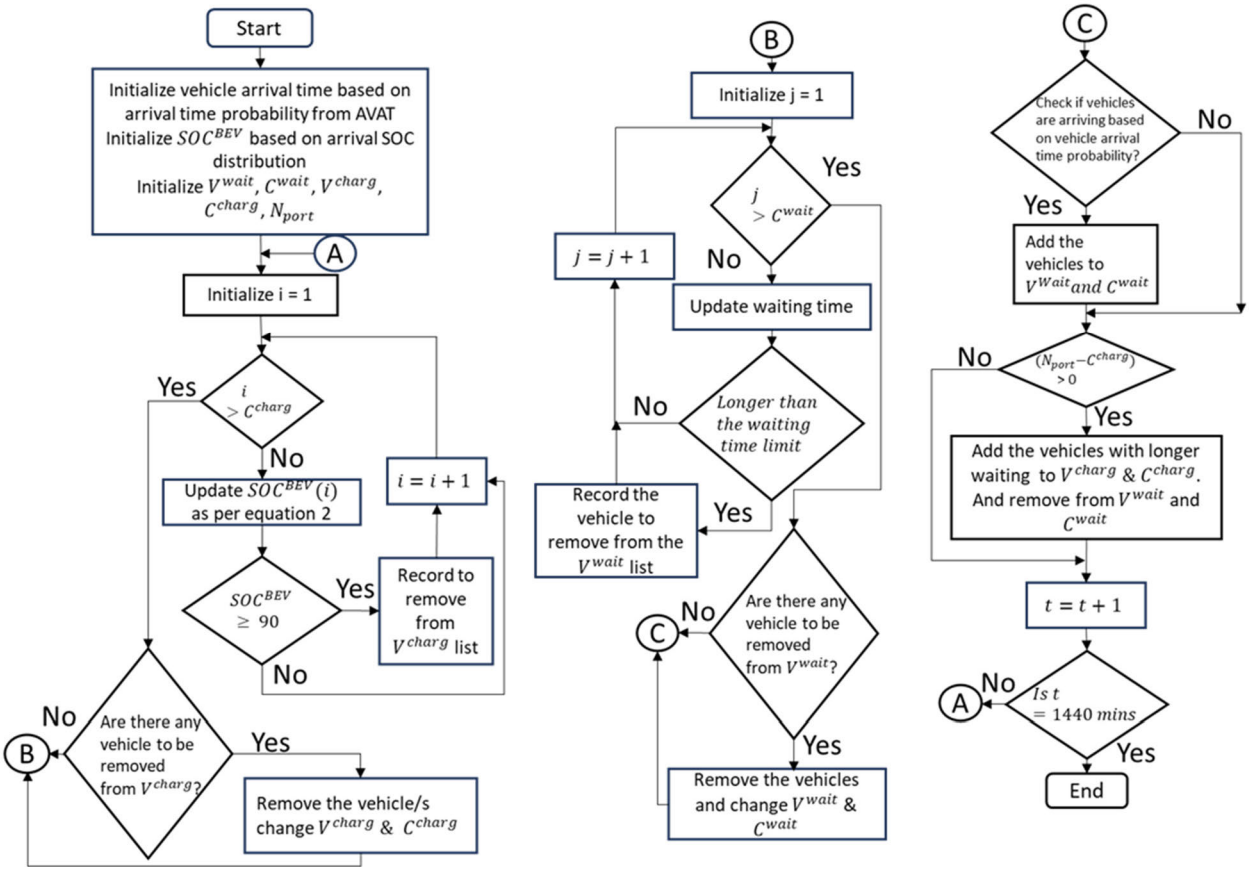


FIGURE 4. Flowchart of AVWT.

charged. A vehicle is added to the charging list from the waiting list or from the new arrivals if there is a charging port available. The SOCs of vehicles in the charging list are updated based on individual characteristics and equation (2). When a vehicle's SOC reaches 90%, then the vehicle is removed from the charging list. At every simulation time instance, both the SOCs of vehicles in the charging list and the waiting times in the waiting list are updated.

Figure 5 summarizes the complete process of estimating vehicle waiting time is shown time at the charging station. In Figure 5, the upper block lists the inputs for the AVAT algorithm to generate the vehicle arrival time probabilities. In the lower block, a list of inputs and the vehicle arrival time probabilities are employed by the AVWT algorithm to generate vehicle waiting time. Thus, the combination of both blocks summarizes how the algorithms are utilized to forecast vehicle waiting time at fast charging stations.

III. MIXED INTEGER DYNAMIC PROGRAMMING FOR CHARGING PLANNING OF AN ELECTRIC VEHICLE CONSIDERING WAITING TIME AT CHARGING STATION

We employ dynamic programming (DP) [38], [39] for optimal charging planning. In the following paragraphs, we discuss the optimization framework for BEV charging planning. The presented DP-based charging plan is distance based. It is assumed that there are multiple charging stations along a route. The locations of charging stations are decision points where the actions of charging/no charging and the amount of charging energy will be determined to minimize the total trip time and charging events and, at the same time, meet the SOE range requirement for the vehicle. The total trip time includes driving time, charging time, and charging waiting time. Table 3 shares the major variables that are considered in the problem formulation. Here, w variables are exogenous inputs to the optimization formulation.

The objective of the optimization is to plan vehicle charging for the whole route and avoid dropping the SOE below manufacturers' recommended limits considered 10% of the battery capacity in this paper. A graphical representation is shown in Figure 6. The cost function for the optimization is defined below:

$$Cost\ Function: \sum_{k=0}^{k=N-1} (T_d(k) + T_{ch}(k) + 1000 * C_{ch}(k)) \tag{3}$$

The objectives are minimizing the total trip time and the number of charging events during the trip. The weighting

TABLE 3. Variables used to describe the optimization framework.

Variables	Description and Units
T_{trip}	Stagewise trip time (min)
T_d	Driving time in the trip (min)
T_{ch}	Time for charging during the trip (min), including charging waiting time
d_s	Travel distance at every stage (meters)
R_{ch}	Charging rate of the charging vehicle (kW) based on the battery characteristics
C_{ch}	Binary variable for charging ($C_{ch} = 1$) not charging ($C_{ch} = 0$) decisions
SOE	State of energy of the vehicle
Max_{charge}	Control variable for limiting the SOE at the charging station
B_{cap}	Battery capacity (kWmin)
w_1	Average link velocity vector ($\frac{m}{s}$)
w_2	Acceleration vector ($\frac{m}{s^2}$)
w_3	Charging station location vector
w_4	Charging waiting time vector (min)
E_{prop}	Propulsion energy (kWmin) consumed at driving the discrete distance d_s

factor 1000 is multiplied to bring the numerical value of the second term equivalent to the time in minutes. Terminal cost is considered a null cost.

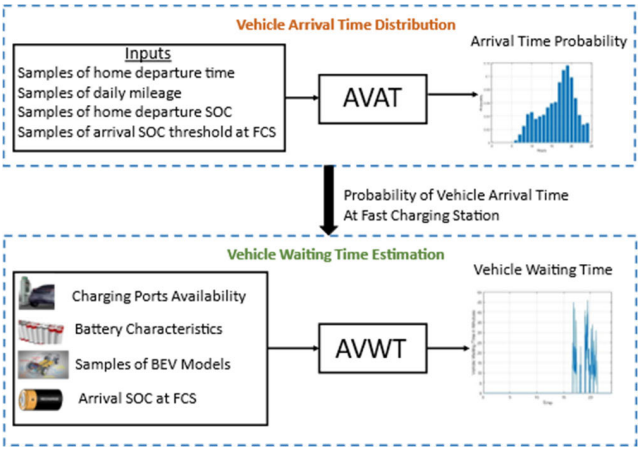


FIGURE 5. Overall framework for forecasting of vehicle waiting time at fasting charging station.

State variables of the optimization are SOE and T_{trip}. SOE is the vehicle's energy status parameter, its fundamental meaning of the parameter is the same as the state of charge. To provide distinction and ease of understanding, we chose to refer to it as SOE here.

$$States : [SOE\ T_{trip}] \tag{4}$$

Control variables of the optimization are charging decision variable C_{ch}, which is binary and SOE charging limit (Max_{charge}). C_{ch} is a binary charging decision variable. Charging variable C_{ch} is 1 when charging station is selected by the optimization algorithm and, 0 when charging the vehicle is not considered. Max_{charge} is the SOE limit to stop charging. The users can specify this value using the app provided by the manufacturer [40].

$$Control\ Variable : [\, C_{ch}\ Max_{charge}\,] \tag{5}$$

Optimization is subjected to dynamics and constraints as defined in the following equations. The change of SOE is described in equation (6). 100 is multiplied to change the ration into percentage.

$$\begin{aligned} SOE\,(k-1) = \Big(& SOE\,(k) - \frac{E_{prop}\,(k)}{B_{cap}} * 100 \\ & + w_3(k) * C_{ch}(k) \\ & * \left(Max_{charge}(k) - SOE\,(k)\right) \Big) \end{aligned} \tag{6}$$

where the energy spent for driving is subtracted, and charging energy is added when charging variable is 1. We consider fast charging stations, however the power accepted is limited by battery charging power characteristics of the specific vehicle type.

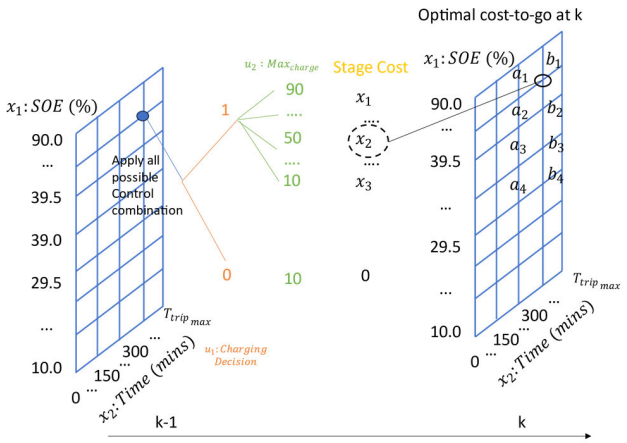


FIGURE 6. Representational computation grid for the DP optimization.

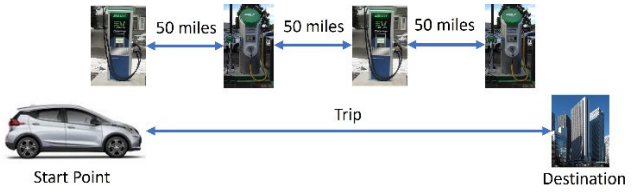


FIGURE 7. Location of charging station on AFC.

Energy spent on propulsion between two charging stations is defined in equation (7) below:

$$E_{prop}(k) = P_{Ev}(k) * \left(\frac{d_s}{w_1(k) * 60}\right) \tag{7}$$

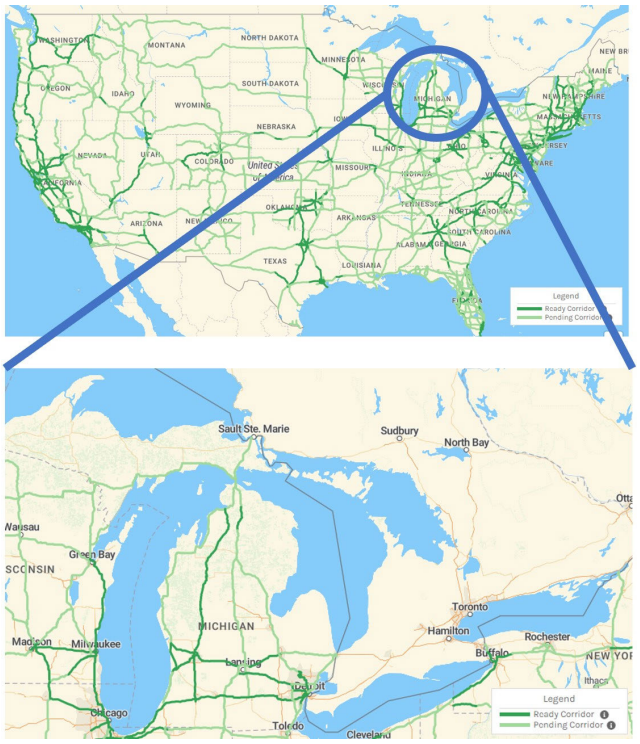


FIGURE 8. Example of alternate fuel corridors [43].

The details of P_{EV} and its variables' computation are provided in equations (8) to (12). Authors in [41] developed this power computation model for GM Bolt 2017. η_{EV} is the variable for efficiency as shown in equation (8) The wheel radius considered is 0.3234 m, and the gear ratio is 7.05 [42]. The road grade (θ) is considered zero. For our study, we chose GM Bolt model 2017. However, other BEV vehicles with different battery technology, vehicle architecture, and vehicle dynamics can have different energy consumption and charging characteristics. Due to this, the optimization framework discussed can output different solutions for charging planning.

$$P_{Ev} = \frac{P_{trac}}{\eta_{EV} * 1000} \tag{8}$$

$$\eta_{EV} = \frac{P_{trac}}{P_{trac} + C_0 + C_1 * w_1 + C_2 * w_1^2 + C_3 * \tau^2} \tag{9}$$

$$P_{trac} = F_t * w_1 \tag{10}$$

$$\tau = (F_t) * \frac{r_w}{gear\ ratio} \tag{11}$$

$$F_t = mass * \left(\alpha + \beta * \sin\theta + \gamma * w_2 + \delta * w_1^2\right) \tag{12}$$

The change is state of time is described in equation

$$T_{trip}\,(k-1) = T_{trip}\,(k) + T_d(k) + T_{ch}(k) \tag{13}$$

$$T_d(k) = \frac{d_s}{w_1(k) * 60} \tag{14}$$

$$T_{ch}\,(k) = \frac{\left(Max_{charge}(k) - SOE\,(k)\right) * B_{cap}}{R_{ch} * 100} + w_4(k) \tag{15}$$

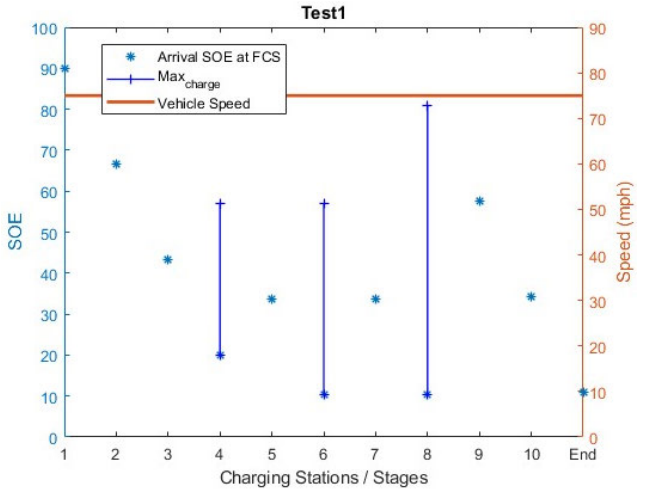


FIGURE 9. SOE status, charging stations and limit selected for test 1.

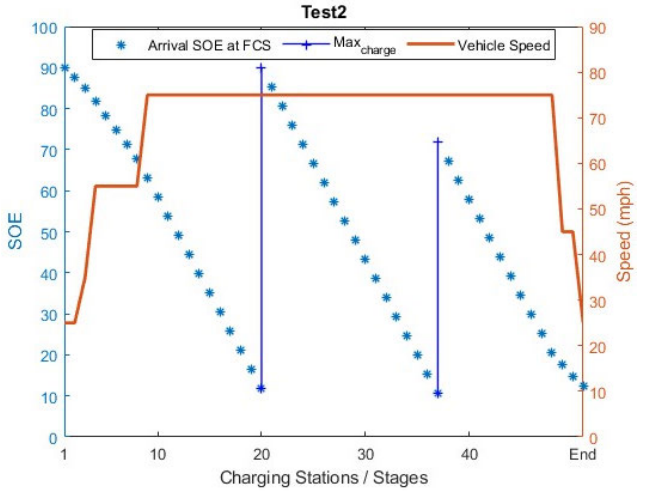


FIGURE 10. SOE status, charging stations and limit selected for test 2.

Equation (14) calculates the driving time, and the value 60 is used to the convert speed unit from m/s to m/min. Equation (15) shows the time equation for calculating the time to charge and the time to wait at the charging station. Table 4 shares the parameters used in the E_{prop} computation in the optimization. The discretization definition mentioned in Table 5 governs the states and Table 6 details the control variables. $T_{trip_{max}}$ is the maximum time limit for the trip.

TABLE 4. Coefficients and constants used in equations.

C_0	1000	C_3	0.00032	δ	0.00012
C_1	10.588	α	0.6633	r_w	0.3234 m
C_2	8.11	β	5813.6	gear ratio	7.05
mass	1616 kg	γ	1.0013		

To avoid mathematical selection of Max_{charge} value below SOE at the current instance k, a penalty cost as described

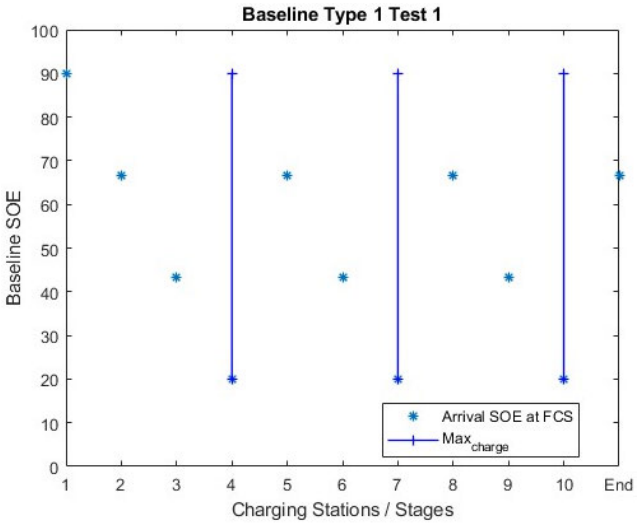


FIGURE 11. SOE status, charging stations and limit for baseline type 1 test 1.

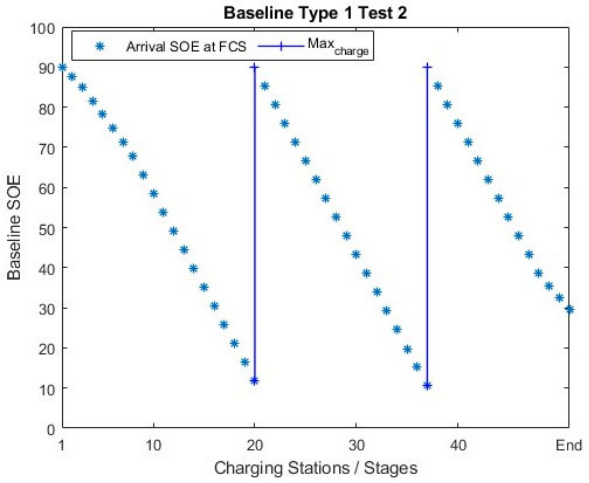


FIGURE 12. SOE status, charging stations and limit for baseline type 1 test 2.

TABLE 5. Discretization of the state variables.

State Variable	Discretization
SOE	10:0.5:90
T_{trip}	0:1:$T_{trip_{max}}$

TABLE 6. Discretization of the control variables.

Control Variable	Discretization
C_{ch}	1
Max_{charge}	10:1:90

in (16) is included.

$$\begin{aligned} &If\ \left(Max_{charge}\left(k\right) - SOE\left(k\right) < 0\right) isTrue, \\ &\quad Then\ add\ penalty \\ &\quad = \left(-100000 * \left(Max_{charge}\left(k\right) - SOE\left(k\right)\right)\right. \end{aligned} \tag{16}$$

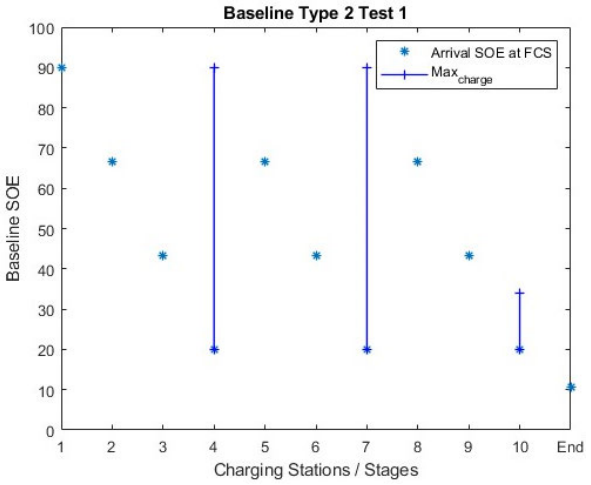


FIGURE 13. SOE status, charging stations and limit for baseline type 2 test 1.

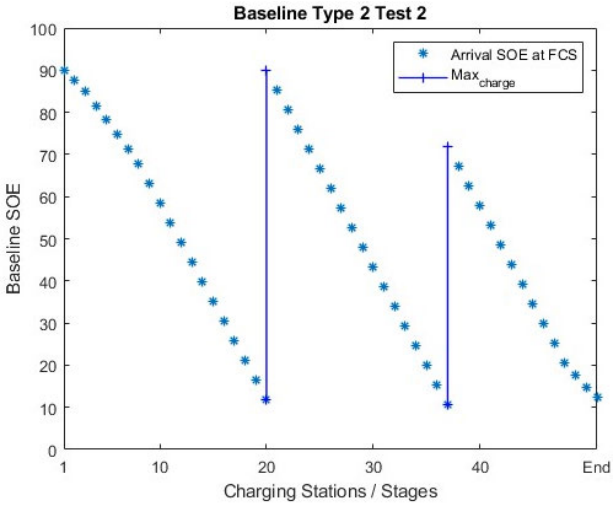


FIGURE 14. SOE status, charging stations and limit for baseline type 2 test 2.

IV. VALIDATION, RESULTS AND DISCUSSION

Typically, long trips are on interstate highways. Governments all over the world are investing in building fast charging stations and encouraging the private sector to invest in creating alternate fuel stations. In the US, one such government initiative is the development of Alternate Fuel Corridors (AFC) [43]; Figure 7 gives a brief overview of a representational trip with the location of charging stations on alternative fuel corridors. All charging stations serve the local population as well as the passing travellers on the route. With the government's support for the adoption of BEV, multiple charging stations using interstate highways are being built all around the Alternative Fuel Corridors. To qualify as an electricity-based fuel corridor, the criteria laid out is that the route should have DC fast charging stations at a gap of every 50 miles on the route and located within a mile range from

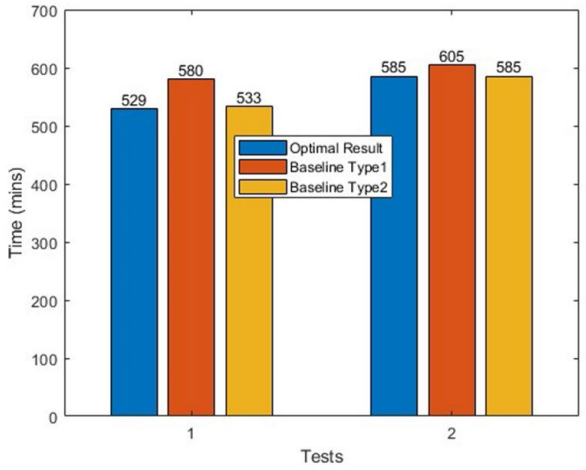


FIGURE 15. Total trip time for optimal results for tests 1 & 2 and baseline types 1 & 2.

the range [44], [45]. Refer to Figure 8 for an AFC network example.

As our test 1, we consider interstate highways where the average link speed is 33.528 m/s (75 mph), and FCS are available at every 80467.2 meters (50 miles). The length of the journey considered is 500 miles. For the described test 1, ds considered is 80467.2 meters (50 miles) in our optimization method. In test 2, we consider a varying link speed during the trip and FCSs are available every 16093.4 meters (10 miles), and ds is 16093.4 meters (10 miles). Since we are considering long trips, we consider the initial SOE as 90%. Hence, the $T_{trip_{max}}$ is considered 540 mins for test case 1 and 600 mins for test case 2. Additional time is given for test 2 as there are many stages that have lower speeds. This will add to the driving time. The start time of the trip is at 8 am as would be the case for a long trip in general.

Figure 9 shows that charging stations 4, 6, and 8 are selected. At charging stations 4 and 6, SOE limit is 57. However, at charging station 8, the SOE limit is 81. The algorithm finishes the trip of test 1 in 529 minutes. The SOE status at the end of the trip is 10.9%. Hence, the residual energy is kept at a minimum. This achieved by only restricting the Max_{charge} limit to keep the residual energy at destination near minimum.

For test 2, as shown in Figure 10, charging stations 20 and 37 were selected. The Max_{charge} selected are 90 and 72 respectively. The time taken to complete the trip for the test condition considered is 585. This is different from test 1 as the average speed is different for some stages in the total trip cumulatively accounting to same trip distance as test 1. The algorithm completes the trip with the remaining SOE of 12.3%.

For comparison, a baseline is considered that the vehicle will be charged when the SOE level is such that it does not have enough energy to reach the next charging station. The maximum charging SOE is 90% for each charging event. This category is referred to as baseline type 1. Figure 11 shows the baseline SOE level of the vehicle and the charging stations selected for baseline type 1 test 1. This approach will have a residual SOE at the end of the trip as 66.6%. The difference between the optimal solution and baseline SOE at the end of the trip is 55.7% for baseline type 1 test 1. This approach has resulted in extra time spent charging to the maximum limit and more than the required energy to reach the destination.

Figure 12 shares the baseline SOE level and charging stations selected for type 1 test 2. Here, the baseline approach will have residual SOE at the end of trip as 29.6%. The difference between optimal solution and baseline SOE at the end of trip is 17.3 %. The optimal solution reaches the destination closer to the lower SOE limit.

When the baseline SOE at the end of the trip is made similar to the end SOE from the optimal solution, the charging limit at the last charging station is reduced to achieve parity. This will make the end SOE level of the optimal solution and baseline similar. This is referred to as baseline type 2. When the rule is applied, the baseline SOE level and charging stations selected for the baseline type 2 test 1 are shown in Figure 13. The Max_{charge} selected at the last charging station is 34%. Figure 14 shows the baseline type 2 test 2 SOE level and charging stations selected for baseline type 2 test 2. The Max_{charge} for baseline type 2 test 2 last charging station is 72%.

Figure 15 shows the trip time for each test and respective baseline. The optimization minimizes trip time compared to the baseline. Table 7 shares the percentage improvement on case-by-case basis. Improvements up to 9.6% are observed.

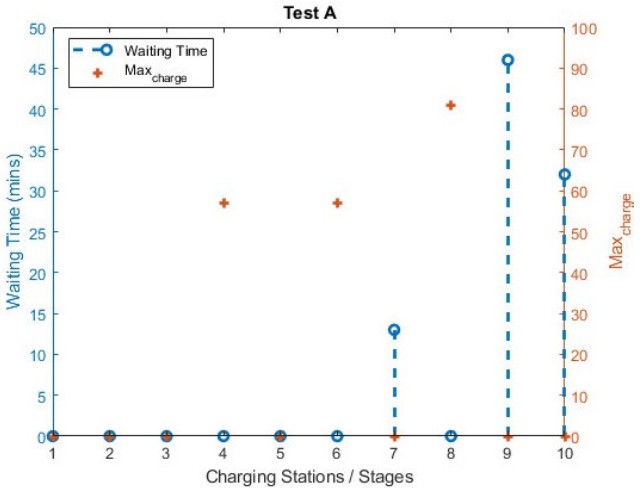


FIGURE 16. Charging stations and limit selected for test A.

In the above cases, waiting times at the charging stations are not encountered due to the start of travel in the early morning, and charging decisions are not made when the waiting time at the fast-charging station is observed. Now, it is considered if the trip starts at noon. Then, a substantial portion of the travel time of the trip will be during the time of the day when waiting time at the charging station will be observed. The tests 1 and 2 conditions, as defined earlier, are

TABLE 7. Improvement for test 1 & test 2 in trip time.

Description	Optimal	Baseline Type 1	Optimal	Baseline Type 2
Test 1 Improvement	9.6%		0.8%	
Test 2 Improvement	3.4%		0.0%	

considered again with a new start time trip at noon. To avoid any confusion, we refer to these modified cases as test A and test B respectively.

Figure 16 shows that the charging stations 4, 6 and 8 are selected. At charging stations 4 and 6, the SOE limit is 57. At the charging station 8 the SOE limit is 81. The algorithm finishes the trip of test 1 in 529 minutes. The SOE status at the end of the trip is 10.9%. The algorithm results avoid waiting time at the charging stations. Figure 17 shows that for Test B, charging stations 20 and 34 were selected. Max_{charge} selected 78 and 86, respectively. The time taken to complete the trip as per the formulation considered is 585. The algorithm completes the trip with the remaining SOE at the end of the trip as 12.3%.

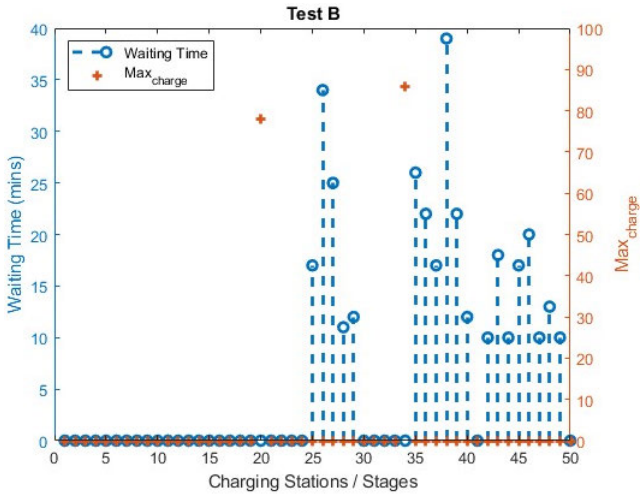


FIGURE 17. Charging stations and limit selected for test B.

We consider the same baseline type 1 and type 2 rules to compare with the optimal solution. The start time is also considered as noon. Figure 18 shares the charging stations and Max_{charge} selected for baseline type 1 test A. The second of the charging stations selected, i.e., seventh of the ten, encounter waiting time at the time of the day the vehicles reach the station. This adds to the total trip time which turns out to be 609 minutes. In Figure 19, it shows that the selected charging stations do not encounter waiting time in baseline type 1 test B. Hence, the total trip time turns out to be 605 minutes.

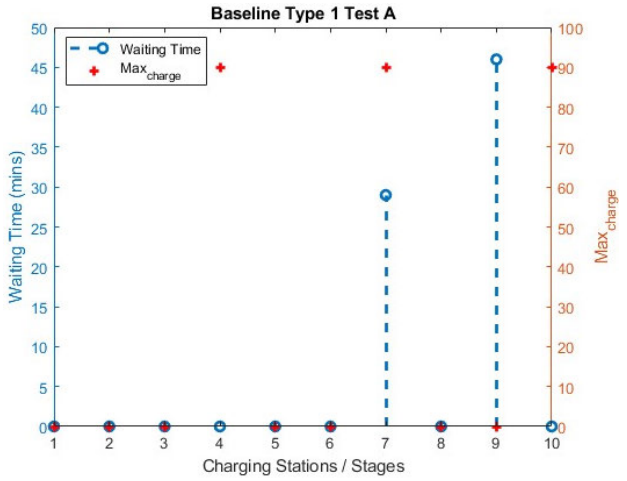


FIGURE 18. Charging stations and limit selected for baseline type 1 test A.

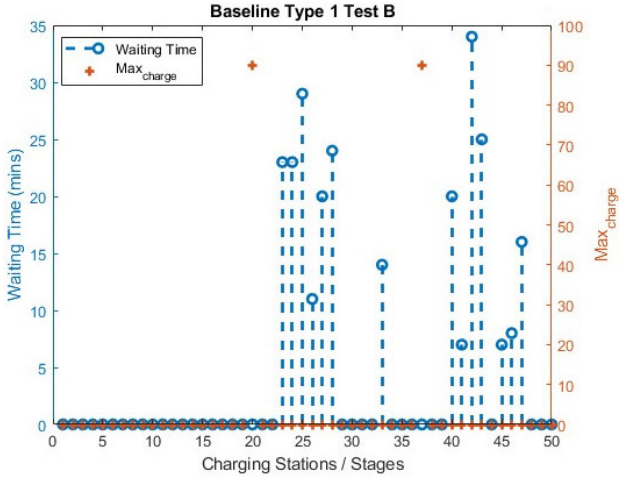


FIGURE 19. Charging stations and limit selected for baseline type 1 test B.

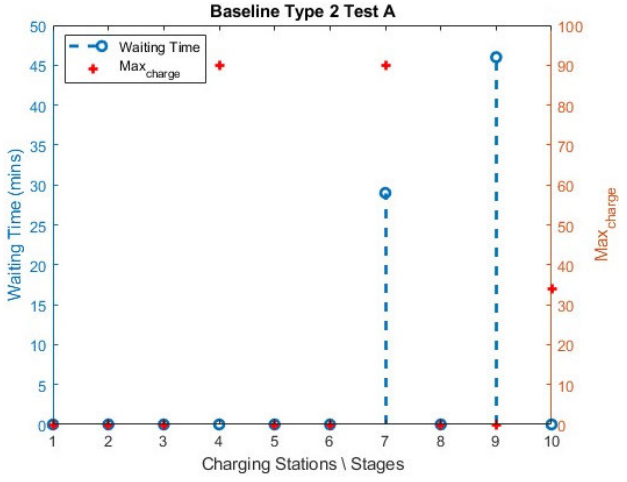


FIGURE 20. Charging stations and limit selected for baseline type 2 test A.

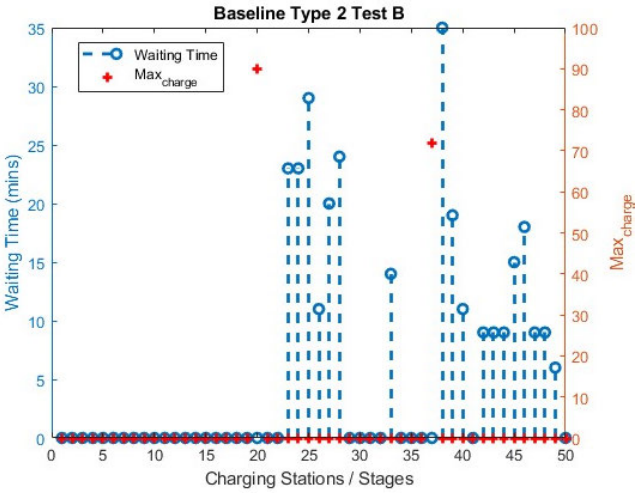


FIGURE 21. Charging stations and limit selected for baseline type 2 test B.

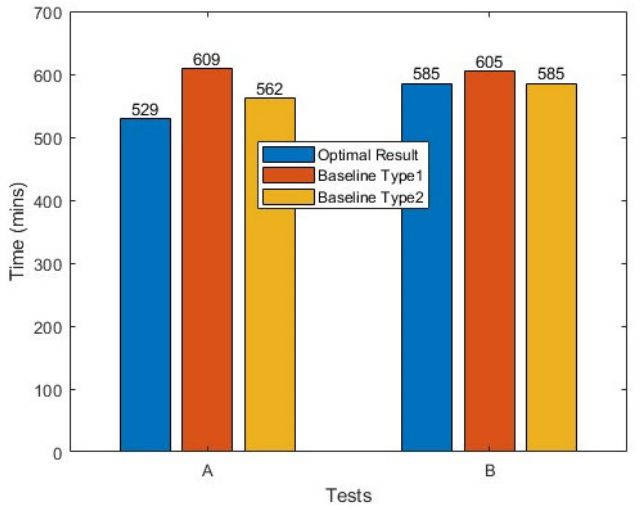


FIGURE 22. Total trip time for optimal results for tests A & B and baseline types 1 & 2.

TABLE 8. Improvement for test A & test B in trip time.

Description	Optimal	Baseline Type 1	Optimal	Baseline Type 2
Test A Improvement	15.1%		6.2%	
Test B Improvement	3.4%		0.0%	

For baseline type 2 test A, based on the rules the total trip time is 562 minutes based on the rules of the baseline type 2. Figure 20 shows the selected charging stations and the Max_{charge} limit for the last charging station reduced to bring it in parity with the end SOE of the optimal solution of test A. Similarly, for baseline type 2 test B, based on the rules of the baseline type 2, the total trip time is 585 minutes. As shown in Figure 21, the selected charging stations remain the same. However, last charging station Max_{charge} is reduced to match the end SOE at the destination about the optimal solution. Waiting time is not encountered in this case as well.

Figure 22 shows the total trip time for all the cases. Table 8 shares the percentage improvements. For test A, due to the waiting time the improvement is considerably more than test B.

V. SUMMARY AND FUTURE WORK

The proposed method in this paper is based on the available public data used for monte-carlo simulation to generate forecasts related to the vehicle waiting time at the fast-charging station. Probabilities of daily trip mileage, daily trip distribution throughout the day and home departure time are utilized to find vehicle arrival time probability at the fast-charging station. A vehicle population with a mix of different battery characteristics and capacities is considered. Dynamic Programming based decision making is employed to observe time savings up to 15.1% with residual SOE at the destination close to a minimum level. This charging planning method significantly enhances the viability of EVs for long-distance travel by addressing core concerns like range anxiety and lengthy charging times. The incorporation of dynamic simulations, probabilistic modeling, and global optimization ensures that drivers can make better-informed decisions on their travel routes. The practical implications range from improving user experience to enabling infrastructure scaling, ultimately pushing for a cleaner, faster, and more efficient future for EVs. By saving time, reducing energy wastage, and improving convenience, this method could be a game-changer for everyday users and long-distance EV travelers.

Below is the future work that can be considered:

1. As more EVs are adopted, consumers' behavior may or may not deviate from its current form. Artificial Intelligence (AI) based methods can be explored as data availability and quality increase.
2. Consumer behavior at the charging station such as tolerance to waiting time and selection of charging at home and/or at FCS, can change based on region and tasks to be accomplished by the vehicle driver.
3. Exploring the implications of varying number of charging ports due to the reasons such as malfunction of EVSE and/or difference in charging station performance. The stochastic nature of charging station availability and incorporating real-time traffic information during the trip can be considered as future work.
4. Apart from Dynamic Programming, AI-based optimal decisions that can incorporate personalized preference for selecting a charging station and charging limit should be explored. The decisions can also be personalized based on driving behavior.
5. Explore possibilities of expanding the solutions of long-trip charging planning in geographical areas with sparse charging infrastructure.

In the future, the BEV adoption is predicted to increase. And the government subsidies die out, the cost of charging may

behave like the current gas cost. Cost will be a very important aspect affecting personal choices.

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